

# **UP Board Class 10 Maths Chapter-wise PYQ (2022–2026)**

यूपी बोर्ड कक्षा 10 गणित – अध्यायवार पिछले वर्षों के प्रश्न

Chapter 5: Arithmetic Progression  
अध्याय 5 : समान्तर श्रेणी

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## अध्याय 5 : समान्तर श्रेणी

## Chapter 5 : Arithmetic Progression

## मुख्य सूत्र / Key Facts

- समान्तर श्रेणी (AP):  $a, a + d, a + 2d, \dots$  जहाँ  $a$  प्रथम पद और  $d$  सार्व अन्तर है;  $d = a_{n+1} - a_n$

Arithmetic Progression (AP):  $a, a + d, a + 2d, \dots$  where  $a$  is the first term and  $d$  the common difference;  $d = a_{n+1} - a_n$

- $n$  वाँ पद:  $a_n = a + (n - 1)d$ ; अन्तिम पद  $l = a + (n - 1)d$

$n$ th term:  $a_n = a + (n - 1)d$ ; last term  $l = a + (n - 1)d$

- प्रथम  $n$  पदों का योग:  $S_n = n/2 [2a + (n - 1)d] = n/2 (a + l)$

Sum of the first  $n$  terms:  $S_n = n/2 [2a + (n - 1)d] = n/2 (a + l)$

- $S_n$  से पद:  $a_n = S_n - S_{n-1}$  ( $n \geq 2$ ) और  $a_1 = S_1$

Term from  $S_n$ :  $a_n = S_n - S_{n-1}$  ( $n \geq 2$ ) and  $a_1 = S_1$

- तीन संख्याएँ  $a, b, c$  AP में हों तो  $2b = a + c$

Three numbers  $a, b, c$  are in AP if  $2b = a + c$

- प्रथम  $n$  प्राकृत संख्याओं का योग  $= n(n + 1)/2$ ; प्रथम  $n$  विषम संख्याओं का योग  $= n^2$

Sum of the first  $n$  natural numbers  $= n(n + 1)/2$ ; sum of the first  $n$  odd numbers  $= n^2$

## बहुविकल्पीय (1 अंक)

## MCQ (1 mark)

(9)

Q1

1 अंक / mark

Common difference

समान्तर श्रेणी (AP)  $-5, -1, 3, 7, \dots$  का सार्व अन्तर होगा :Common difference for the Arithmetic Progression (AP)  $-5, -1, 3, 7, \dots$  will be :

- A. 1  
B. 2  
C. 3  
D. 4

Asked in: 2024 (Set HV, Q.5)

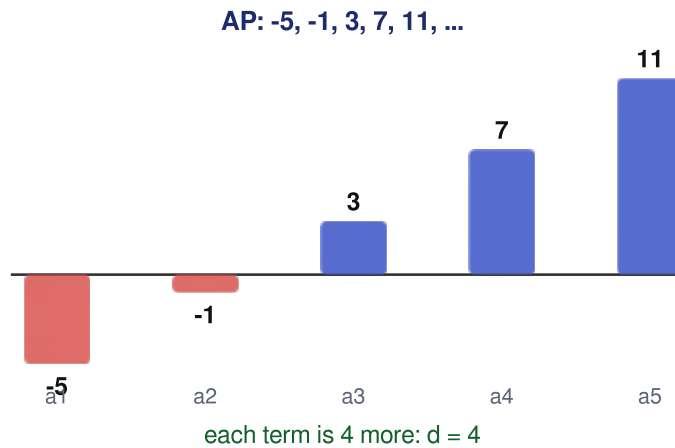
✓ सही उत्तर / Correct answer: (D)

## हल (हिन्दी)

- सार्व अन्तर  $d =$  दूसरा पद  $-$  पहला पद  $= -1 - (-5) = 4$
- जाँच:  $3 - (-1) = 4$  और  $7 - 3 = 4$  ✓
- उत्तर: (D) 4।

## Solution (English)

- Common difference  $d =$  second term  $-$  first term  $= -1 - (-5) = 4$
- Check:  $3 - (-1) = 4$  and  $7 - 3 = 4$  ✓
- Answer: (D) 4.

Fig: 2024 - HV - Q.5 (solution diagram) - terms of the AP  $-5, -1, 3, 7, \dots$  with common difference 4

Q2

1 अंक / mark

Sum of n terms

पहले 15 गुणजों (8 के) का योगफल होगा :

The sum of first 15 multiples of 8 will be :

- A. 960  
B. 980  
C. 984  
D. 990

Asked in: 2024 (Set HX, Q.5)

✓ सही उत्तर / Correct answer: (A)

हल (हिन्दी)

- 8 के पहले 15 गुणज: 8, 16, 24, ..., 120 — यह AP है जिसमें  $a = 8$ ,  $d = 8$ ,  $n = 15$
- $S_n = n/2 [2a + (n - 1)d] = 15/2 [16 + 14 \times 8] = 15/2 \times 128 = 15 \times 64 = 960$
- उत्तर: (A) 960 ।

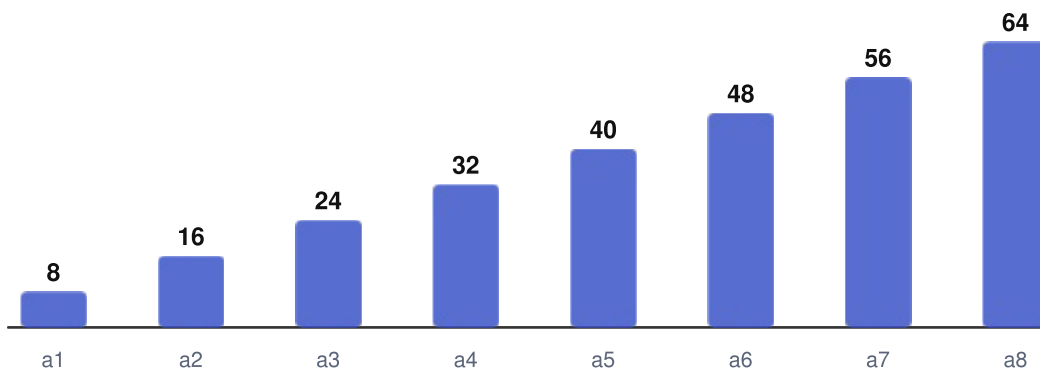
Solution (English)

- The first 15 multiples of 8: 8, 16, 24, ..., 120 — an AP with  $a = 8$ ,  $d = 8$ ,  $n = 15$
- $S_n = n/2 [2a + (n - 1)d] = 15/2 [16 + 14 \times 8] = 15/2 \times 128 = 15 \times 64 = 960$
- Answer: (A) 960.

नोट: हिन्दी पाठ स्कैन में नहीं देखा गया; अंग्रेज़ी प्रश्न के अनुसार हल दिया गया है ।

Note: the Hindi wording was not read in the scan; the solution follows the English question.

First multiples of 8: 8, 16, 24, ...



$$S_{15} = 15/2 \times (8 + 120) = 960$$

Fig: 2024 - HX - Q.5 (solution diagram) - multiples of 8:  $a = 8$ ,  $d = 8$ , 15 terms with sum 960

Q3 1 अंक / mark nth term

अनुक्रम  $\sqrt{2}, 2\sqrt{2}, 3\sqrt{2}, \dots$  का 10वाँ पद होगा :10th term of the sequence  $\sqrt{2}, 2\sqrt{2}, 3\sqrt{2}, \dots$  will be :

- A.  $\sqrt{242}$   
 B.  $\sqrt{288}$   
 C.  $\sqrt{200}$   
 D.  $\sqrt{162}$

Asked in: 2024 (Set HZ, Q.5)

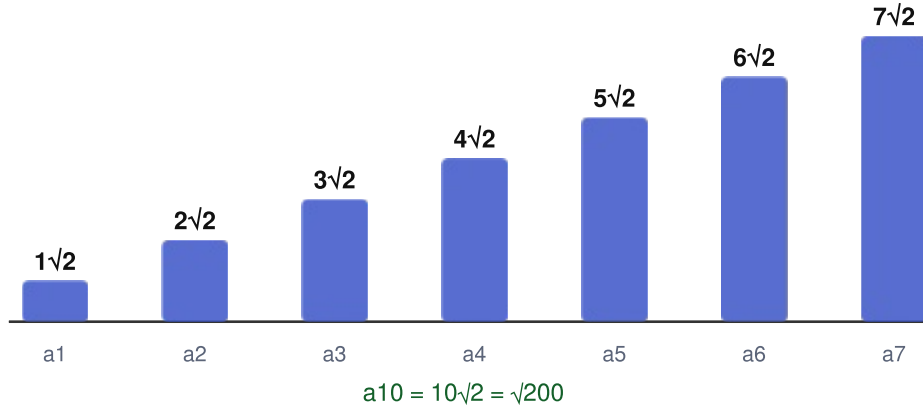
✓ सही उत्तर / Correct answer: (C)

हल (हिन्दी)

- यह AP है:  $a = \sqrt{2}$ ,  $d = 2\sqrt{2} - \sqrt{2} = \sqrt{2}$
- $a_{10} = a + 9d = \sqrt{2} + 9\sqrt{2} = 10\sqrt{2}$
- $10\sqrt{2} = \sqrt{(100 \times 2)} = \sqrt{200}$  | उत्तर: (C)  $\sqrt{200}$  |

Solution (English)

- This is an AP:  $a = \sqrt{2}$ ,  $d = 2\sqrt{2} - \sqrt{2} = \sqrt{2}$
- $a_{10} = a + 9d = \sqrt{2} + 9\sqrt{2} = 10\sqrt{2}$
- $10\sqrt{2} = \sqrt{(100 \times 2)} = \sqrt{200}$ . Answer: (C)  $\sqrt{200}$ .

Sequence  $\sqrt{2}, 2\sqrt{2}, 3\sqrt{2}, \dots$ Fig: 2024 - HZ - Q.5 (solution diagram) - terms  $k\sqrt{2}$  grow by  $\sqrt{2}$  each time; the 10th term is  $10\sqrt{2} = \sqrt{200}$

**Q4** 1 अंक / mark nth term

समान्तर श्रेणी 10, 7, 4, ... का 20वाँ पद होगा :

20th term of the A.P. 10, 7, 4, ... will be :

- A. -57  
 B. 57  
**C. -47**  
 D. 47

Asked in: 2025 (Set CA, Q.12)

✓ सही उत्तर / Correct answer: (C)

हल (हिन्दी)

- $a = 10, d = 7 - 10 = -3, n = 20$
- $a_n = a + (n - 1)d \Rightarrow a_{20} = 10 + 19 \times (-3) = 10 - 57 = -47$
- उत्तर: (C) -47।

Solution (English)

- $a = 10, d = 7 - 10 = -3, n = 20$
- $a_n = a + (n - 1)d \Rightarrow a_{20} = 10 + 19 \times (-3) = 10 - 57 = -47$
- Answer: (C) -47.

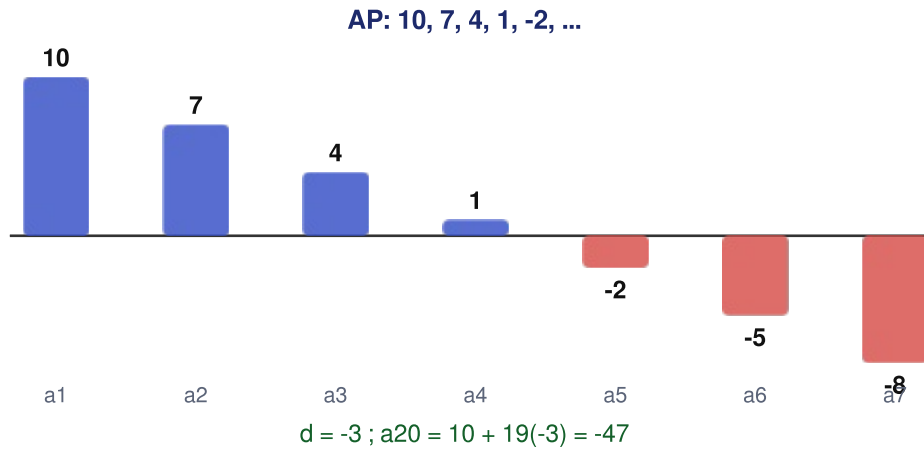


Fig: 2025 - CA - Q.12 (solution diagram) - terms of 10, 7, 4, ... decreasing by 3; the 20th term is -47

**Q5** 1 अंक / mark nth term**A.P. 6, 10, 14, ... का 25वाँ पद होगा :****25th term of an A.P. 6, 10, 14, ... will be :**

- A. 100  
**B. 102**  
 C. 101  
 D. 151

Asked in: 2025 (Set BY, Q.6)

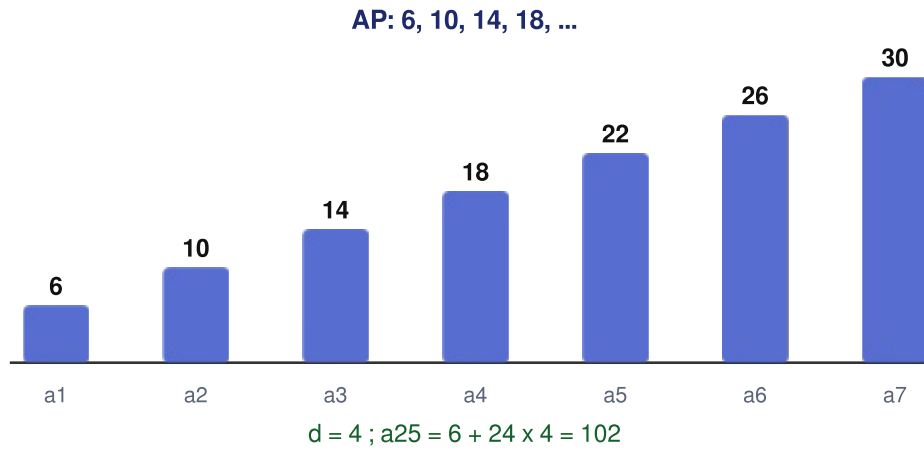
✓ सही उत्तर / Correct answer: (B)

**हल (हिन्दी)**

- $a = 6, d = 10 - 6 = 4, n = 25$
- $a_{25} = a + 24d = 6 + 24 \times 4 = 6 + 96 = 102$
- उत्तर: (B) 102।

**Solution (English)**

- $a = 6, d = 10 - 6 = 4, n = 25$
- $a_{25} = a + 24d = 6 + 24 \times 4 = 6 + 96 = 102$
- Answer: (B) 102.

Fig: 2025 – BY – Q.6 (solution diagram) – terms of 6, 10, 14, ... with  $d = 4$ ; the 25th term is 102

**Q6** 1 अंक / mark nth term**A.P.  $-6, -4, -2, \dots$  का सातवाँ पद होगा :****The seventh term of A.P.  $-6, -4, -2, \dots$  will be :**

- A.  $-8$   
 B.  $18$   
 C.  $8$   
 D.  $6$

Asked in: 2025 (Set BW, Q.20)

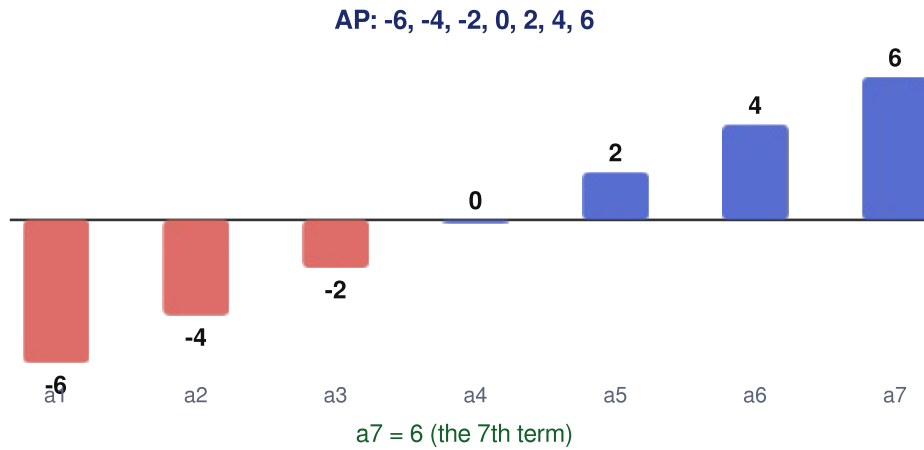
✓ सही उत्तर / Correct answer: (D)

**हल (हिन्दी)**

- $a = -6, d = -4 - (-6) = 2, n = 7$
- $a_7 = a + 6d = -6 + 6 \times 2 = -6 + 12 = 6$
- उत्तर: (D) 6।

**Solution (English)**

- $a = -6, d = -4 - (-6) = 2, n = 7$
- $a_7 = a + 6d = -6 + 6 \times 2 = -6 + 12 = 6$
- Answer: (D) 6.

Fig: 2025 – BW – Q.20 (solution diagram) – the first seven terms of  $-6, -4, -2, \dots$ ; the 7th term is 6

Q7

1 अंक / mark

Finding n

समान्तर श्रेणी 21, 18, 15, ... का वह पद जिसका मान  $-81$  है, होगा :

The value of a term of an A.P. 21, 18, 15, ... is  $-81$ . Then the term will be :

- A. 31वाँ / 31st  
 B. 35वाँ / 35th  
 C. 33वाँ / 33rd  
 D. 37वाँ / 37th

Asked in: 2026 (Set GA, Q.14)

✓ सही उत्तर / Correct answer: (B)

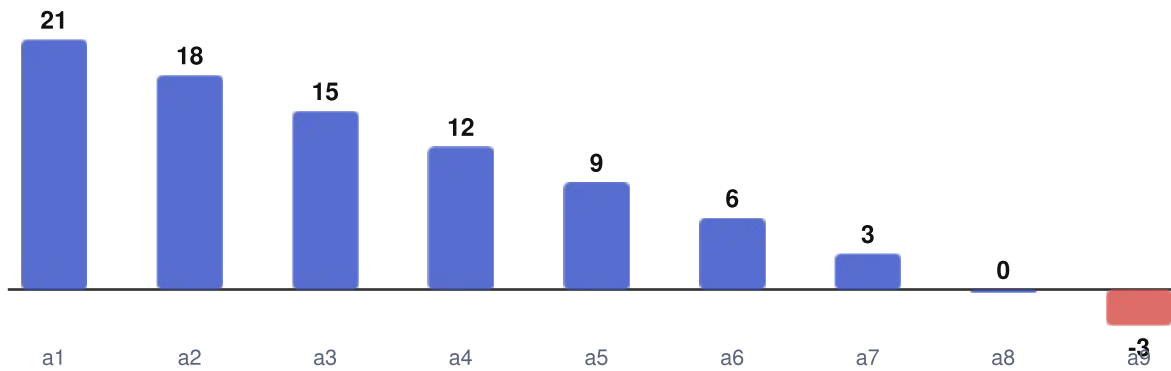
हल (हिन्दी)

- $a = 21, d = 18 - 21 = -3, a_n = -81$
- $a + (n - 1)d = -81 \Rightarrow 21 - 3(n - 1) = -81 \Rightarrow -3(n - 1) = -102$
- $n - 1 = 34 \Rightarrow n = 35$
- उत्तर: (B) 35वाँ पद।

Solution (English)

- $a = 21, d = 18 - 21 = -3, a_n = -81$
- $a + (n - 1)d = -81 \Rightarrow 21 - 3(n - 1) = -81 \Rightarrow -3(n - 1) = -102$
- $n - 1 = 34 \Rightarrow n = 35$
- Answer: (B) 35th term.

AP: 21, 18, 15, ... (first 9 terms)



$$21 - 3(n - 1) = -81 \Rightarrow n = 35$$

Fig: 2026 - GA - Q.14 (solution diagram) - terms of 21, 18, 15, ... ; the value  $-81$  is reached at the 35th term

Q8 1 अंक / mark nth term

**A.P.  $3/2, 1/2, -1/2, \dots$  का 10वाँ पद होगा :****The 10th term of the A.P.  $3/2, 1/2, -1/2, \dots$  will be :**

- A.  $-15/2$   
 B.  $-13/2$   
 C.  $-11/2$   
 D.  $-9/2$

Asked in: 2026 (Set GB, Q.6)

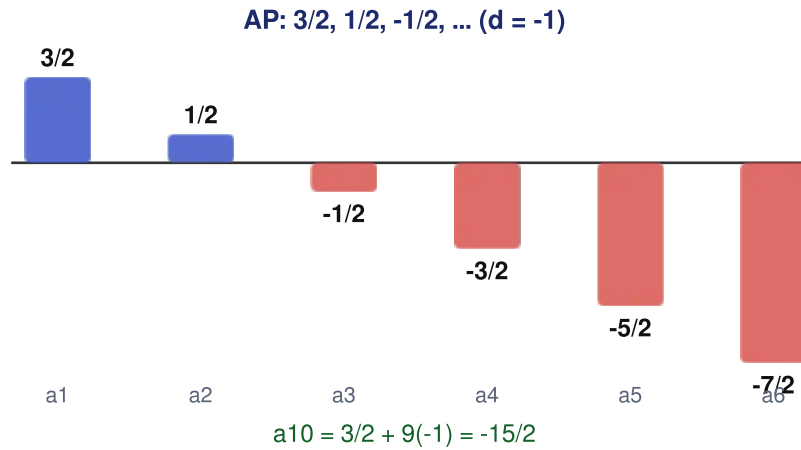
✓ सही उत्तर / Correct answer: (A)

**हल (हिन्दी)**

- $a = 3/2, d = 1/2 - 3/2 = -1, n = 10$
- $a_{10} = a + 9d = 3/2 + 9(-1) = 3/2 - 9 = (3 - 18)/2 = -15/2$
- उत्तर: (A)  $-15/2$ ।

**Solution (English)**

- $a = 3/2, d = 1/2 - 3/2 = -1, n = 10$
- $a_{10} = a + 9d = 3/2 + 9(-1) = 3/2 - 9 = (3 - 18)/2 = -15/2$
- Answer: (A)  $-15/2$ .

Fig: 2026 - GB - Q.6 (solution diagram) - terms of  $3/2, 1/2, -1/2, \dots$  ; the 10th term is  $-15/2$

Q9

1 अंक / mark

nth term

**A.P. : 3, 8, 13, ... का सत्रहवाँ पद होगा :****The seventeenth term of the A.P. : 3, 8, 13, ... will be :****A. 83**

B. 88

C. 98

D. 105

Asked in: 2026 (Set FX, Q.5)

✓ सही उत्तर / Correct answer: (A)

**हल (हिन्दी)**

1.  $a = 3, d = 8 - 3 = 5, n = 17$

2.  $a_{17} = a + 16d = 3 + 16 \times 5 = 3 + 80 = 83$

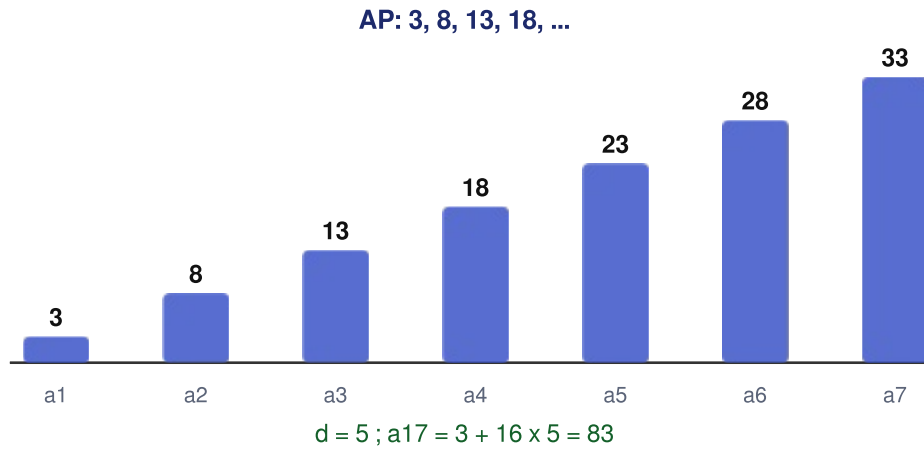
3. उत्तर: (A) 83।

**Solution (English)**

1.  $a = 3, d = 8 - 3 = 5, n = 17$

2.  $a_{17} = a + 16d = 3 + 16 \times 5 = 3 + 80 = 83$

3. Answer: (A) 83.

Fig: 2026 – FX – Q.5 (solution diagram) – terms of 3, 8, 13, ... with  $d = 5$ ; the 17th term is 83

## लघु उत्तरीय (2 अंक)

## Short answer (2 marks)

(1)

Q10

2 अंक / marks

Condition for AP

**x के किस मान के लिए  $2x$ ,  $x + 8$  तथा  $3x + 1$  समान्तर श्रेणी में होंगे ?****For what value of x will  $2x$ ,  $x + 8$  and  $3x + 1$  be in arithmetic progression ?**

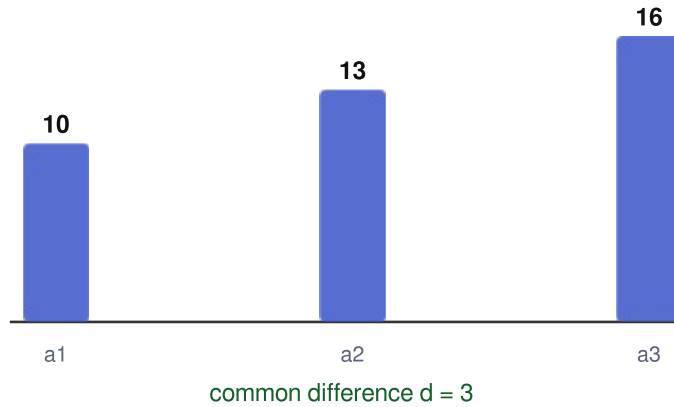
Asked in: 2024 (Set IB, Q.1(b))

## हल (हिन्दी)

1. तीन संख्याएँ  $a$ ,  $b$ ,  $c$  समान्तर श्रेणी में हों तो  $2b = a + c$
2. यहाँ  $a = 2x$ ,  $b = x + 8$ ,  $c = 3x + 1$
3.  $2(x + 8) = 2x + (3x + 1) \Rightarrow 2x + 16 = 5x + 1$
4.  $16 - 1 = 5x - 2x \Rightarrow 3x = 15 \Rightarrow x = 5$
5. जाँच:  $x = 5$  पर पद 10, 13, 16 हैं;  $13 - 10 = 3$  और  $16 - 13 = 3$  ✓
6. अतः  $x = 5$

## Solution (English)

1. If three numbers  $a$ ,  $b$ ,  $c$  are in AP, then  $2b = a + c$
2. Here  $a = 2x$ ,  $b = x + 8$ ,  $c = 3x + 1$
3.  $2(x + 8) = 2x + (3x + 1) \Rightarrow 2x + 16 = 5x + 1$
4.  $16 - 1 = 5x - 2x \Rightarrow 3x = 15 \Rightarrow x = 5$
5. Check: for  $x = 5$  the terms are 10, 13, 16;  $13 - 10 = 3$  and  $16 - 13 = 3$  ✓
6. Hence  $x = 5$

 **$x = 5 : 2x, x + 8, 3x + 1 = 10, 13, 16$** Fig: 2024 - IB - Q.1(b) (solution diagram) - for  $x = 5$  the three terms are 10, 13, 16 ( $d = 3$ )

## दीर्घ उत्तरीय (4 अंक)

## Long answer (4 marks)

(5)

Q11 4 अंक / marks Sum of n terms

A.P. 21, 18, 15, ... के कितने पद लेने चाहिए कि उनका योग  $-81$  हो ?How many terms of the A.P. : 21, 18, 15, ... must be taken so that their sum is  $-81$  ?

Asked in: 2025 (Set BW, Q.2(c))

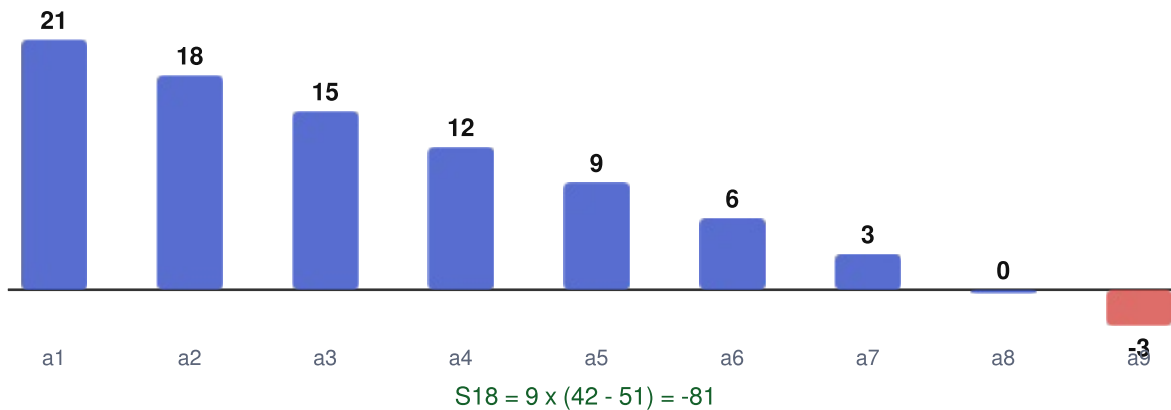
## हल (हिन्दी)

1.  $a = 21, d = 18 - 21 = -3, S_n = -81$
2.  $S_n = n/2 [2a + (n - 1)d] \Rightarrow -81 = n/2 [42 - 3(n - 1)] = n/2 (45 - 3n)$
3.  $-162 = 45n - 3n^2 \Rightarrow 3n^2 - 45n - 162 = 0 \Rightarrow n^2 - 15n - 54 = 0$
4.  $(n - 18)(n + 3) = 0 \Rightarrow n = 18$  या  $n = -3$
5.  $n$  ऋणात्मक नहीं हो सकता, अतः  $n = 18$
6. जाँच:  $S_{18} = 9 [42 - 51] = 9 \times (-9) = -81 \checkmark$
7. अतः 18 पद लेने होंगे।

## Solution (English)

1.  $a = 21, d = 18 - 21 = -3, S_n = -81$
2.  $S_n = n/2 [2a + (n - 1)d] \Rightarrow -81 = n/2 [42 - 3(n - 1)] = n/2 (45 - 3n)$
3.  $-162 = 45n - 3n^2 \Rightarrow 3n^2 - 45n - 162 = 0 \Rightarrow n^2 - 15n - 54 = 0$
4.  $(n - 18)(n + 3) = 0 \Rightarrow n = 18$  or  $n = -3$
5.  $n$  cannot be negative, so  $n = 18$
6. Check:  $S_{18} = 9 [42 - 51] = 9 \times (-9) = -81 \checkmark$
7. So 18 terms must be taken.

AP: 21, 18, 15, ..., 0, -3 (first 9 terms)

Fig: 2025 - BW - Q.2(c) (solution diagram) - terms of 21, 18, 15, ... ; the sum of the first 18 terms is  $-81$ 

Q12 4 अंक / marks Common difference

किसी A.P. का 17वाँ पद उसके 10वें पद से 7 अधिक है। सार्व अन्तर ज्ञात कीजिए।

The 17th term of an A.P. exceeds its 10th term by 7. Find the common difference.

Asked in: 2026 (Set GA, Q.2(d))

## हल (हिन्दी)

1. दिया है:  $a_{17} - a_{10} = 7$
2.  $(a + 16d) - (a + 9d) = 7$
3.  $7d = 7 \Rightarrow d = 1$
4. अतः सार्व अन्तर  $d = 1$

## Solution (English)

1. Given:  $a_{17} - a_{10} = 7$
2.  $(a + 16d) - (a + 9d) = 7$
3.  $7d = 7 \Rightarrow d = 1$
4. So the common difference  $d = 1$

Q13

4 अंक / marks

Sum of n terms

**A.P. 8, 3, -2, ... के प्रथम 21 पदों का योग ज्ञात कीजिए।****Find the sum of first 21 terms of the A.P. 8, 3, -2, ... .**

Asked in: 2026 (Set GB, Q.2(f))

**हल (हिन्दी)**

1.  $a = 8, d = 3 - 8 = -5, n = 21$
2.  $S_n = n/2 [2a + (n - 1)d] \Rightarrow S_{21} = 21/2 [16 + 20 \times (-5)]$
3.  $= 21/2 [16 - 100] = 21/2 \times (-84) = 21 \times (-42)$
4.  $S_{21} = -882$

**Solution (English)**

1.  $a = 8, d = 3 - 8 = -5, n = 21$
2.  $S_n = n/2 [2a + (n - 1)d] \Rightarrow S_{21} = 21/2 [16 + 20 \times (-5)]$
3.  $= 21/2 [16 - 100] = 21/2 \times (-84) = 21 \times (-42)$
4.  $S_{21} = -882$

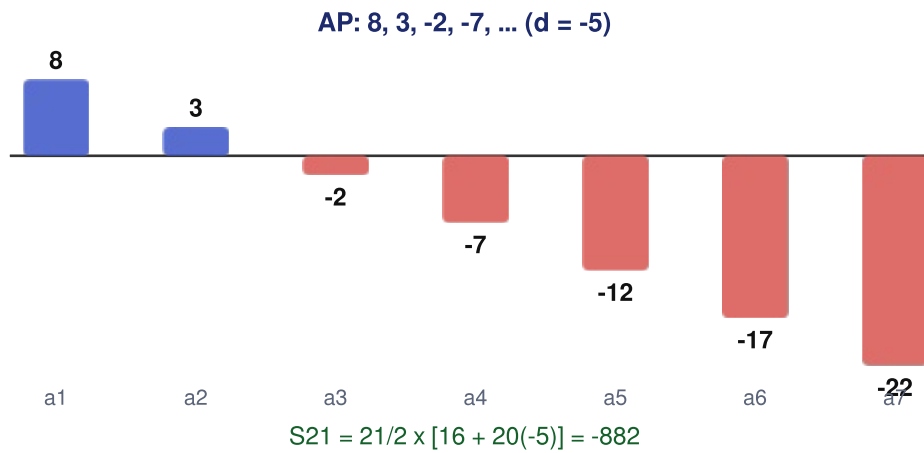


Fig: 2026 - GB - Q.2(f) (solution diagram) - terms of 8, 3, -2, ... ; the sum of the first 21 terms is -882

Q14

4 अंक / marks

nth term from  $S_n$ 

किसी समान्तर श्रेणी के प्रथम  $n$  पदों का योगफल  $3n^2 + 4n$  है। इस समान्तर श्रेणी का 25वाँ पद ज्ञात कीजिए।

The sum of first  $n$  terms of an A.P. is  $3n^2 + 4n$ . Find the 25th term of the Arithmetic Progression (A.P.).

Asked in: 2026 (Set FW, Q.2(a))

हल (हिन्दी)

1. दिया है:  $S_n = 3n^2 + 4n$
2.  $a_{25} = S_{25} - S_{24}$
3.  $S_{25} = 3(625) + 4(25) = 1875 + 100 = 1975$
4.  $S_{24} = 3(576) + 4(24) = 1728 + 96 = 1824$
5.  $a_{25} = 1975 - 1824 = 151$
6. जाँच:  $a_n = S_n - S_{n-1} = 6n + 1 \Rightarrow a_{25} = 6 \times 25 + 1 = 151 \checkmark$
7. अतः 25वाँ पद 151 है।

Solution (English)

1. Given:  $S_n = 3n^2 + 4n$
2.  $a_{25} = S_{25} - S_{24}$
3.  $S_{25} = 3(625) + 4(25) = 1875 + 100 = 1975$
4.  $S_{24} = 3(576) + 4(24) = 1728 + 96 = 1824$
5.  $a_{25} = 1975 - 1824 = 151$
6. Check:  $a_n = S_n - S_{n-1} = 6n + 1 \Rightarrow a_{25} = 6 \times 25 + 1 = 151 \checkmark$
7. So the 25th term is 151.

नोट: स्कैन में  $S_n$  की अभिव्यक्ति  $3n^2 + 4n$  पढ़ी गई (अंग्रेजी पाठ में " $3n^2 + 4n$ ")।

Note: the expression for  $S_n$  is read as  $3n^2 + 4n$  in the scan.

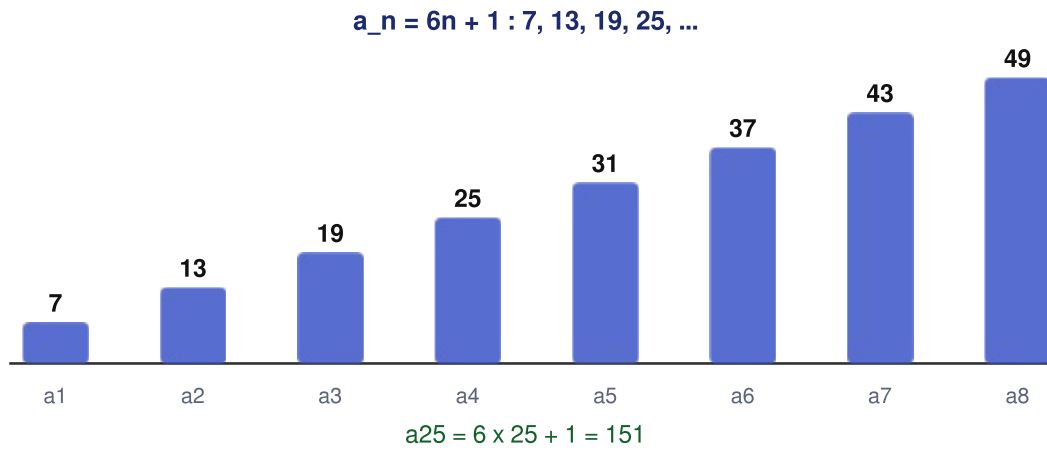


Fig: 2026 - FW - Q.2(a) (solution diagram) - the AP 7, 13, 19, ... ( $S_n = 3n^2 + 4n$ ,  $a_n = 6n + 1$ ); the 25th term is 151

**Q15**

4 अंक / marks

Number of terms and sum

किसी A.P. में प्रथम पद और अन्तिम पद क्रमशः 8 और 341 हैं। यदि सार्व अन्तर 9 है, तो इसमें कुल पदों की संख्या तथा उनका योग ज्ञात कीजिए।

The first term and the last term of an A.P. are 8 and 341 respectively. If the common difference is 9, then find the number of terms in it and their sum.

Asked in: 2026 (Set FX, Q.2(e))

**हल (हिन्दी)**

1.  $a = 8, l = 341, d = 9$
2.  $l = a + (n - 1)d \Rightarrow 341 = 8 + (n - 1) \times 9 \Rightarrow 333 = 9(n - 1) \Rightarrow n - 1 = 37 \Rightarrow n = 38$
3. योग  $S_n = n/2 (a + l) = 38/2 \times (8 + 341) = 19 \times 349 = 6631$
4. अतः पदों की संख्या 38 और उनका योग 6631 है।

**Solution (English)**

1.  $a = 8, l = 341, d = 9$
2.  $l = a + (n - 1)d \Rightarrow 341 = 8 + (n - 1) \times 9 \Rightarrow 333 = 9(n - 1) \Rightarrow n - 1 = 37 \Rightarrow n = 38$
3. Sum  $S_n = n/2 (a + l) = 38/2 \times (8 + 341) = 19 \times 349 = 6631$
4. So the number of terms is 38 and their sum is 6631.

**AP: 8, 17, 26, 35, ... , 341**

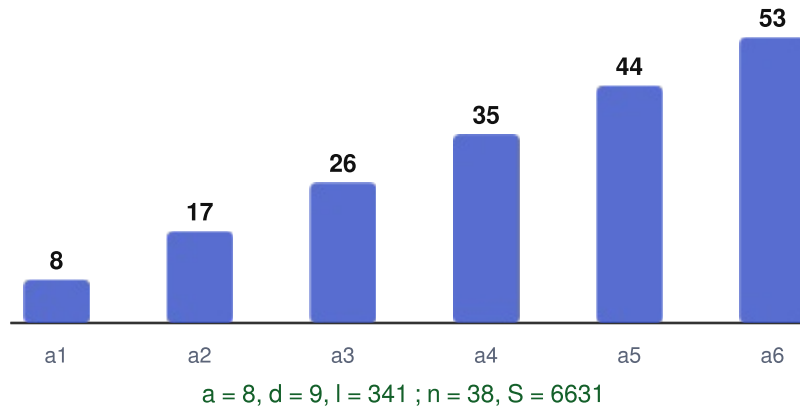


Fig: 2026 - FX - Q.2(e) (solution diagram) - terms of 8, 17, 26, ..., 341: there are 38 terms in all

## दीर्घ उत्तरीय (6 अंक)

## Long answer (6 marks)

(4)

Q16

6 अंक / marks

Number of terms and common difference

किसी समान्तर श्रेणी का प्रथम पद 5, अन्तिम पद 64 और सभी पदों का योगफल 400 है। पदों की संख्या एवं सार्व अन्तर ज्ञात कीजिए।

First term of an A.P. is 5, last term is 64 and sum of all the terms is 400. Find the number of terms and common difference.

Asked in: 2025 (Set CA, Q.3)

## हल (हिन्दी)

1. दिया है:  $a = 5$ ,  $l = 64$ ,  $S_n = 400$
2.  $S_n = n/2 (a + l) \Rightarrow 400 = n/2 \times (5 + 64) = n/2 \times 69$
3.  $n = 800/69 \approx 11.6$  — यह प्राकृत संख्या नहीं है, इसलिए छपे आँकड़ों (अन्तिम पद 64) के साथ ऐसी कोई A.P. सम्भव नहीं है।
4. सम्भावित सही रूप: यदि अन्तिम पद 45 होता (NCERT का मानक प्रश्न), तो  $400 = n/2 \times (5 + 45) = 25n \Rightarrow n = 16$
5.  $l = a + (n - 1)d \Rightarrow 45 = 5 + 15d \Rightarrow d = 40/15 = 8/3$
6. उत्तर (अन्तिम पद 45 मानने पर): पदों की संख्या 16 और सार्व अन्तर 8/3।
7. छपे अनुसार (64):  $n = 800/69$  पूर्णांक नहीं है, इसलिए यह प्रश्न छपाई-त्रुटि वाला लगता है।

## Solution (English)

1. Given:  $a = 5$ ,  $l = 64$ ,  $S_n = 400$
2.  $S_n = n/2 (a + l) \Rightarrow 400 = n/2 \times (5 + 64) = n/2 \times 69$
3.  $n = 800/69 \approx 11.6$  — not a natural number, so no AP exists with the printed data (last term 64).
4. Probable intended form: if the last term were 45 (the standard NCERT question),  $400 = n/2 \times (5 + 45) = 25n \Rightarrow n = 16$
5.  $l = a + (n - 1)d \Rightarrow 45 = 5 + 15d \Rightarrow d = 40/15 = 8/3$
6. Answer (taking the last term as 45): 16 terms and common difference 8/3.
7. As printed (64):  $n = 800/69$  is not an integer, so this question seems to contain a printing error.

नोट: हिन्दी और अंग्रेज़ी दोनों पाठों में अन्तिम पद "64" छपा दिखता है (स्कैन पर मुहर का निशान है)। इससे  $n$  पूर्णांक नहीं आता, इसलिए हल दोनों रूपों में दिया गया है। इसे "Unclear" सूची में भी रखा गया है।

Note: the last term appears as "64" in both the Hindi and English texts (a stamp overlaps it in the scan). That makes  $n$  a non-integer, so the solution is given for both readings. It is also listed under "Unclear".

Q17

6 अंक / marks

Sum of n terms

संख्याओं की उस सूची के प्रथम 24 पदों का योग ज्ञात कीजिए जिसका  $n$  वाँ पद  $a_n = 3 + 2n$  से दिया जाता है।

Find the sum of first 24 terms of the list of numbers whose  $n$ -th term is given by  $a_n = 3 + 2n$ .

Asked in: 2025 (Set CB, Q.3)

हल (हिन्दी)

1.  $a_n = 3 + 2n$  से:  $a_1 = 5, a_2 = 7, a_3 = 9, \dots$
2.  $a_2 - a_1 = 7 - 5 = 2, a_3 - a_2 = 2 \Rightarrow$  सूची एक AP है जिसमें  $a = 5, d = 2$
3.  $a_{24} = 3 + 2 \times 24 = 51$
4.  $S_{24} = n/2 (a + l) = 24/2 \times (5 + 51) = 12 \times 56 = 672$
5. जाँच:  $S_{24} = 24/2 [10 + 23 \times 2] = 12 \times 56 = 672 \checkmark$
6. अतः प्रथम 24 पदों का योग 672 है।

Solution (English)

1. From  $a_n = 3 + 2n$ :  $a_1 = 5, a_2 = 7, a_3 = 9, \dots$
2.  $a_2 - a_1 = 7 - 5 = 2, a_3 - a_2 = 2 \Rightarrow$  the list is an AP with  $a = 5, d = 2$
3.  $a_{24} = 3 + 2 \times 24 = 51$
4.  $S_{24} = n/2 (a + l) = 24/2 \times (5 + 51) = 12 \times 56 = 672$
5. Check:  $S_{24} = 24/2 [10 + 23 \times 2] = 12 \times 56 = 672 \checkmark$
6. So the sum of the first 24 terms is 672.

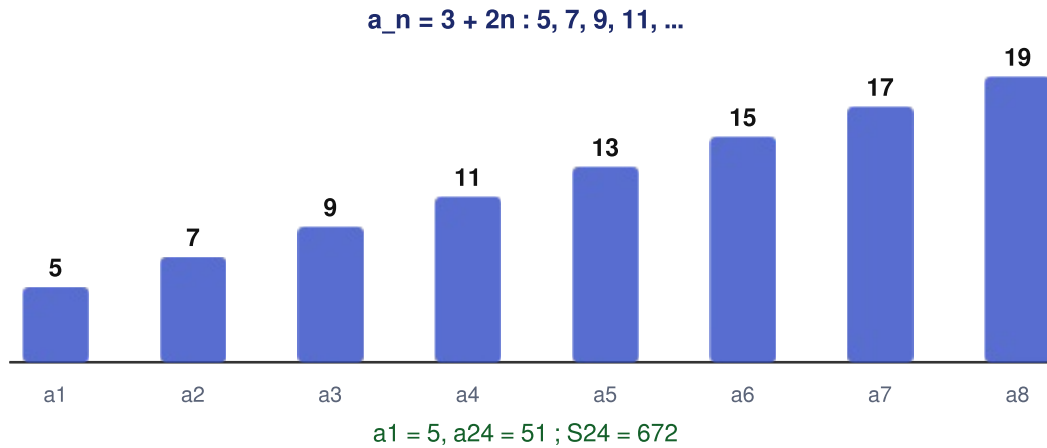


Fig: 2025 - CB - Q.3 (solution diagram) - the list 5, 7, 9, ... ( $a_n = 3 + 2n$ ); the sum of its first 24 terms is 672

Q18

6 अंक / marks

Sum of n terms

यदि किसी A.P. के प्रथम 7 पदों का योग 49 और प्रथम 17 पदों का योग 289 है, तो इसके प्रथम  $n$  पदों का योग ज्ञात कीजिए।

If the sum of first 7 terms of an A.P. is 49 and the sum of first 17 terms is 289, find the sum of first  $n$  terms.

Asked in: 2025 (Set BY, Q.3)

हल (हिन्दी)

- माना प्रथम पद  $a$  और सार्व अन्तर  $d$  है।
- $S_7 = 7/2 [2a + 6d] = 49 \Rightarrow 2a + 6d = 14 \Rightarrow a + 3d = 7 \dots (i)$
- $S_{17} = 17/2 [2a + 16d] = 289 \Rightarrow 17(a + 8d) = 289 \Rightarrow a + 8d = 17 \dots (ii)$
- $(ii) - (i): 5d = 10 \Rightarrow d = 2$ ; (i) से  $a = 7 - 6 = 1$
- $S_n = n/2 [2a + (n - 1)d] = n/2 [2 + 2(n - 1)] = n/2 \times 2n = n^2$
- जाँच:  $S_7 = 49 \checkmark$ ,  $S_{17} = 289 \checkmark$
- अतः  $S_n = n^2$

Solution (English)

- Let the first term be  $a$  and the common difference  $d$ .
- $S_7 = 7/2 [2a + 6d] = 49 \Rightarrow 2a + 6d = 14 \Rightarrow a + 3d = 7 \dots (i)$
- $S_{17} = 17/2 [2a + 16d] = 289 \Rightarrow 17(a + 8d) = 289 \Rightarrow a + 8d = 17 \dots (ii)$
- $(ii) - (i): 5d = 10 \Rightarrow d = 2$ ; from (i)  $a = 7 - 6 = 1$
- $S_n = n/2 [2a + (n - 1)d] = n/2 [2 + 2(n - 1)] = n/2 \times 2n = n^2$
- Check:  $S_7 = 49 \checkmark$ ,  $S_{17} = 289 \checkmark$
- So  $S_n = n^2$

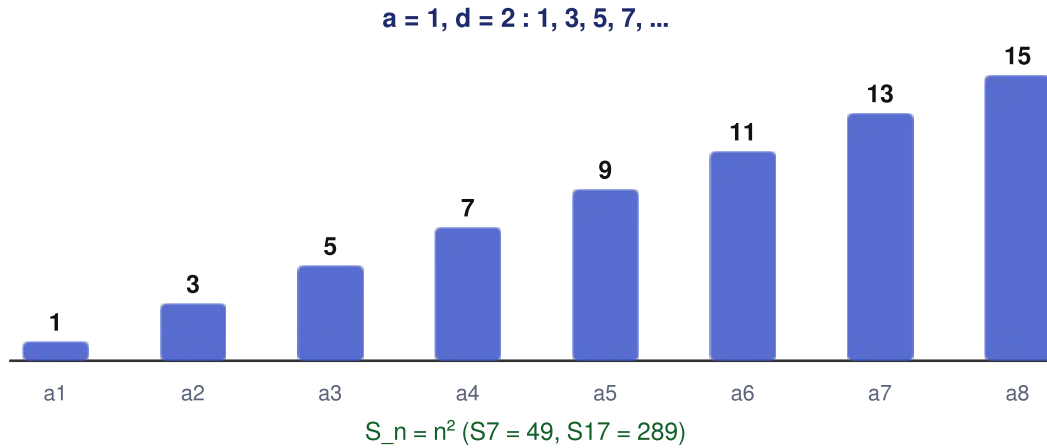


Fig: 2025 - BY - Q.3 (solution diagram) - the AP 1, 3, 5, 7, ... has  $S_7 = 49$  and  $S_{17} = 289$ , so  $S_n = n^2$

Q19

6 अंक / marks

Word problem (repaying a loan)

एक आदमी ने ₹ 3,250 का ऋण लिया। उसने पहले महीने ₹ 20 दिए, फिर वह प्रति माह ₹ 15 बढ़ाता गया। तो वह कितने माह में ऋण चुका देगा ?

A man borrowed ₹ 3,250. He paid ₹ 20 in the first month and then went on increasing his payment by ₹ 15 every month. In how many months will he repay the loan ?

Asked in: 2024 (Set HZ, Q.3 OR)

हल (हिन्दी)

- मासिक किश्तें 20, 35, 50, 65, ... एक AP है:  $a = 20$ ,  $d = 15$ ;  $S_n = 3250$
- $S_n = n/2 [2a + (n-1)d] \Rightarrow 3250 = n/2 [40 + 15(n-1)] = n/2 (15n + 25)$
- $6500 = 15n^2 + 25n \Rightarrow 3n^2 + 5n - 1300 = 0$
- $D = 25 + 4 \times 3 \times 1300 = 25 + 15600 = 15625 = 125^2 \Rightarrow n = (-5 \pm 125)/6$
- $n = 120/6 = 20$  या  $n = -130/6$  (अस्वीकार)
- जाँच:  $S_{20} = 10 [40 + 285] = 10 \times 325 = 3250$  ✓
- अतः वह 20 माह में ऋण चुका देगा।

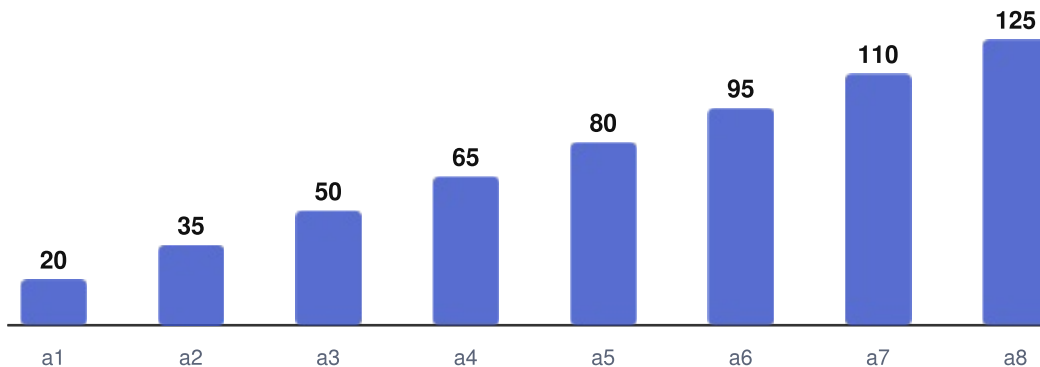
Solution (English)

- The monthly payments 20, 35, 50, 65, ... form an AP:  $a = 20$ ,  $d = 15$ ;  $S_n = 3250$
- $S_n = n/2 [2a + (n-1)d] \Rightarrow 3250 = n/2 [40 + 15(n-1)] = n/2 (15n + 25)$
- $6500 = 15n^2 + 25n \Rightarrow 3n^2 + 5n - 1300 = 0$
- $D = 25 + 4 \times 3 \times 1300 = 25 + 15600 = 15625 = 125^2 \Rightarrow n = (-5 \pm 125)/6$
- $n = 120/6 = 20$  or  $n = -130/6$  (rejected)
- Check:  $S_{20} = 10 [40 + 285] = 10 \times 325 = 3250$  ✓
- So he will repay the loan in 20 months.**

नोट: प्रश्न "अथवा" वाला विकल्प था (दूसरा विकल्प रैखिक समीकरण पर आधारित था)।

Note: this was the "OR" alternative of the question (the other alternative was a linear-equation problem).

Monthly payments: 20, 35, 50, 65, ...



20 months :  $S_{20} = 10 \times (40 + 285) = 3250$

Fig: 2024 - HZ - Q.3 OR (solution diagram) - monthly payments 20, 35, 50, ... ( $d = 15$ ); 20 payments total ₹ 3250